

How big is the home-court advantage in the NBA? A season-by-season analysis of 11 731 games, 2016–17 to 2025–26

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Introduction

Home advantage is one of the most discussed effects in sport. In basketball it can be explained by a combination of factors: a supportive crowd, the away team being tired from travelling, familiarity with the home arena, and possibly referees being influenced by the crowd. Because every NBA game is recorded, the effect can be quantified rather than debated about. In this report I ask how large the home-court advantage in the NBA was between the 2016–17 and 2025–26 seasons, whether it is statistically significant, and whether it changes from season to season. Without any home advantage the home team would win half of its games, so 50% is the reference value throughout.

Data sample

I used a public table of NBA game results on Kaggle [1] covering ten seasons. Seasons are labelled by the year in which they end, so 2017 is the 2016–17 season and 2026 is the 2025–26 season. Each row is one game, with the date, season, home and away team, a flag for neutral-site games, whether the home team won, and the point margin (home minus away points). The raw file contains 11 777 games. Two kinds of games were removed: games at a neutral venue, where neither team is really at home, and All-Star weekend exhibition games, which use invented team codes and are not competitive NBA games. Keeping only games between two of the 30 NBA franchises leaves **11 731 games**. The analysis was written in Python (pandas, NumPy, matplotlib) in Visual Studio Code.

Method

For each season i the home-win rate is $r_i = W_i/N_i$, with W_i the home wins and N_i the games played, and the overall rate is $P_{\text{home}} = \sum W_i / \sum N_i$. To estimate the statistical uncertainty, each season is treated as an independent sub-sample ($K = 10$), and the uncertainty is the sample standard deviation of the seasonal rates,

$$s = \sqrt{\frac{1}{K-1}\sum_{i=1}^{K}(r_i - \bar{r})^2}. \tag{1}$$

Using seasons rather than random sub-samples has the advantage that the spread also shows directly whether the rate moves over time. The significance of the home advantage is $z = (P_{\text{home}} - 0.50)/s$, and I take $z > 3$ as significant. A possible trend is measured with the Pearson correlation coefficient r between season and r_i, tested with $t = r\sqrt{K-2}/\sqrt{1-r^2}$ at the 5% level (8 degrees of freedom). Finally, the distribution of the point margin is plotted and its mean calculated.

Results and discussion

Over all 11 731 games the home team won 6 593, a home-win rate of 56.20%. Table 1 gives the result for each of the ten seasons.

The home team won more than half of its games in every season, with rates between 53.86% (2021) and 59.17% (2019). The standard deviation of the ten seasonal rates is $s = 2.02$ percentage points, so

$$P_{\text{home}} = (56.20 \pm 2.02)\%, \qquad z = \frac{0.5620 - 0.50}{0.0202} = 3.06\,\sigma. \tag{2}$$

The home advantage is therefore statistically significant. Taking the uncertainty of the mean instead, $s/\sqrt{10} = 0.64$ points, would give $z = 9.7\sigma$, so the conclusion does not depend on this choice; s is the more conservative of the two. Figure 1 shows the same seasonal rates graphically.

Table 1: Games, home wins and home-win rate per season after cleaning (season 2017 = 2016–17).

Season	Games	Home wins	Home-win rate (%)
2017	1 227	716	58.35
2018	1 227	711	57.95
2019	1 227	726	59.17
2020	968	535	55.27
2021	945	509	53.86
2022	1 230	669	54.39
2023	1 229	714	58.10
2024	1 228	667	54.32
2025	1 225	667	54.45
2026	1 225	679	55.43
Total	**11 731**	**6 593**	**56.20**

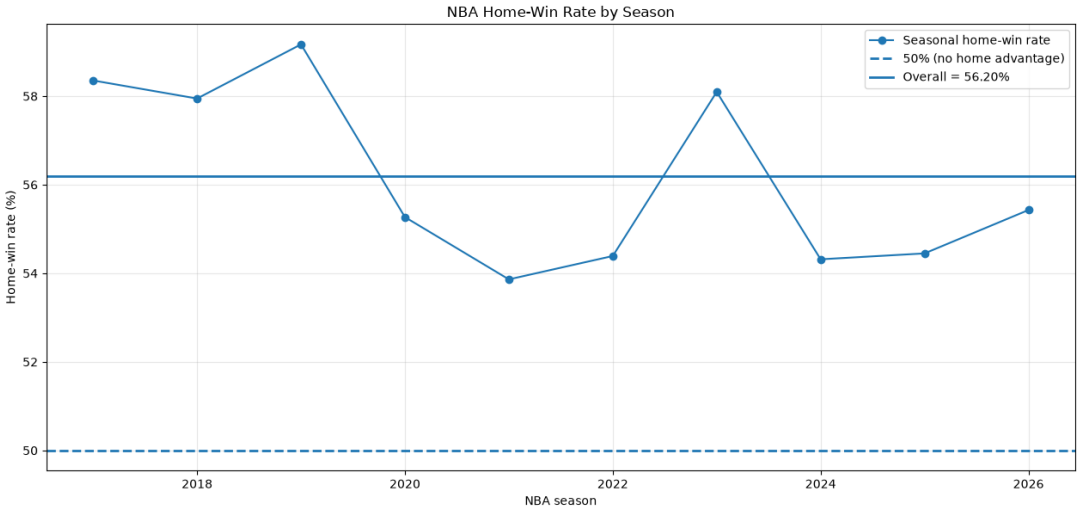


Figure 1: Home-win rate per season. Dashed line: 50% (no home advantage); solid horizontal line: overall rate of 56.20%.

The figure also shows that the rate is not constant. The first three seasons are all close to 58–59%, while most later seasons are around 54–55%, with 2023 as the one clear exception. The correlation between season and home-win rate is $r = -0.599$, but the t-test gives $|t| = 2.12$, below the critical value of 2.31 ($p \approx 0.07$), so this downward trend is not significant with only ten seasons. The two lowest-volume seasons, 2020 and 2021, were played with few or no fans because of COVID-19 and are among the lowest in the table, which contributes to the negative trend. That the home team still won 53.86% in 2021 suggests that the advantage is not entirely due to the home crowd.

The point margin shows the same effect under an other angle. Figure 2 shows its distribution over all 11 731 games. It is broad, with most games decided by fewer than 20 points, but it is clearly shifted to positive values: on average the home team scores 2.12 points more than the away team.

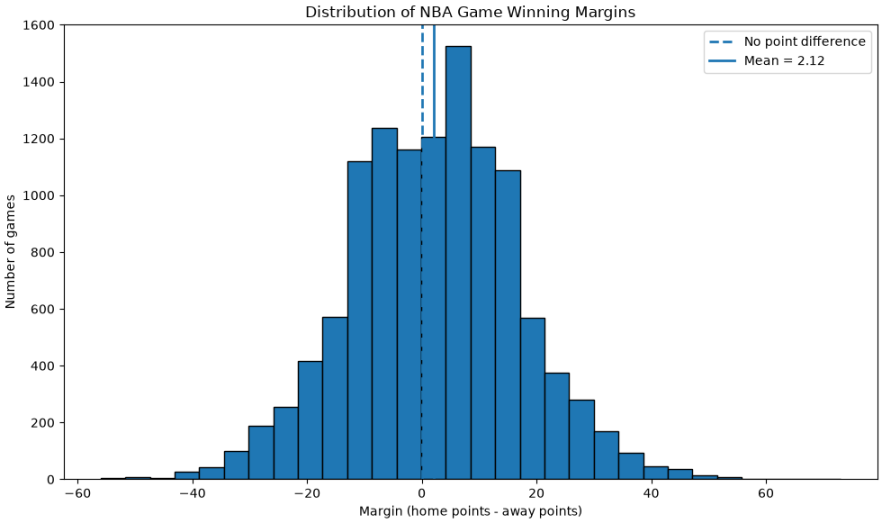


Figure 2: Distribution of the point margin (home minus away points) for all 11 731 games. Dashed line: no point difference; solid line: the mean margin of +2.12 points.

Conclusion

There is a statistically significant home-court advantage in the NBA. Between 2016–17 and 2025–26 the home team won $(56.20 \pm 2.02)\%$ of its games, 3.06σ above 50%, and won by +2.12 points on average. The rate appears to decrease over the period ($r = -0.599$), but this trend is not significant. The main limitations are the small number of sub-samples (ten seasons), the two pandemic-affected seasons, and the fact that the data contain only results, without information on travel, back-to-back games, time zones or rested players. The size of the effect can therefore be measured, but not which of these causes matters most.

References

[1] **Author: MrBridge**. *NBA Home-Court Advantage 2017-2026: 11,777 Games*. Available at: **https://www.kaggle.com/datasets/mrbridge/nba-home-court-2017-2026**.